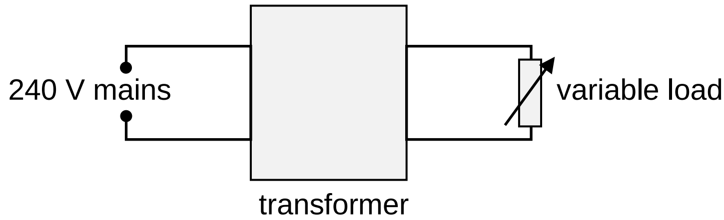
27 An ideal transformer steps down the 240 V sinusoidal voltage from the mains to 12 V, which is then applied to a variable load.



What is the change in the current supplied by the mains when the resistance of the load is increased from 20 Ω to 50 Ω?

A Decrease from 30 mA to 12 mA.

B Increase from 12 mA to 30 mA.

C Decrease from 12 mA to 4.8 mA.

D Increase from 0.24 A to 0.60 A.

Ans: A

$I_P / I_S = n_S / n_P = V_S / V_P = 12 / 240 = 0.050$

At 20 Ω, $I_S = 12 / 20 = 0.60$ A $\rightarrow$ $I_P = 0.60 \times 0.050 = 0.030$ A

At 50 Ω, $I_S = 12 / 50 = 0.24$ A $\rightarrow$ $I_P = 0.24 \times 0.050 = 0.012$ A

28 The diagram shows two spectra of X-rays from an X-ray tube.